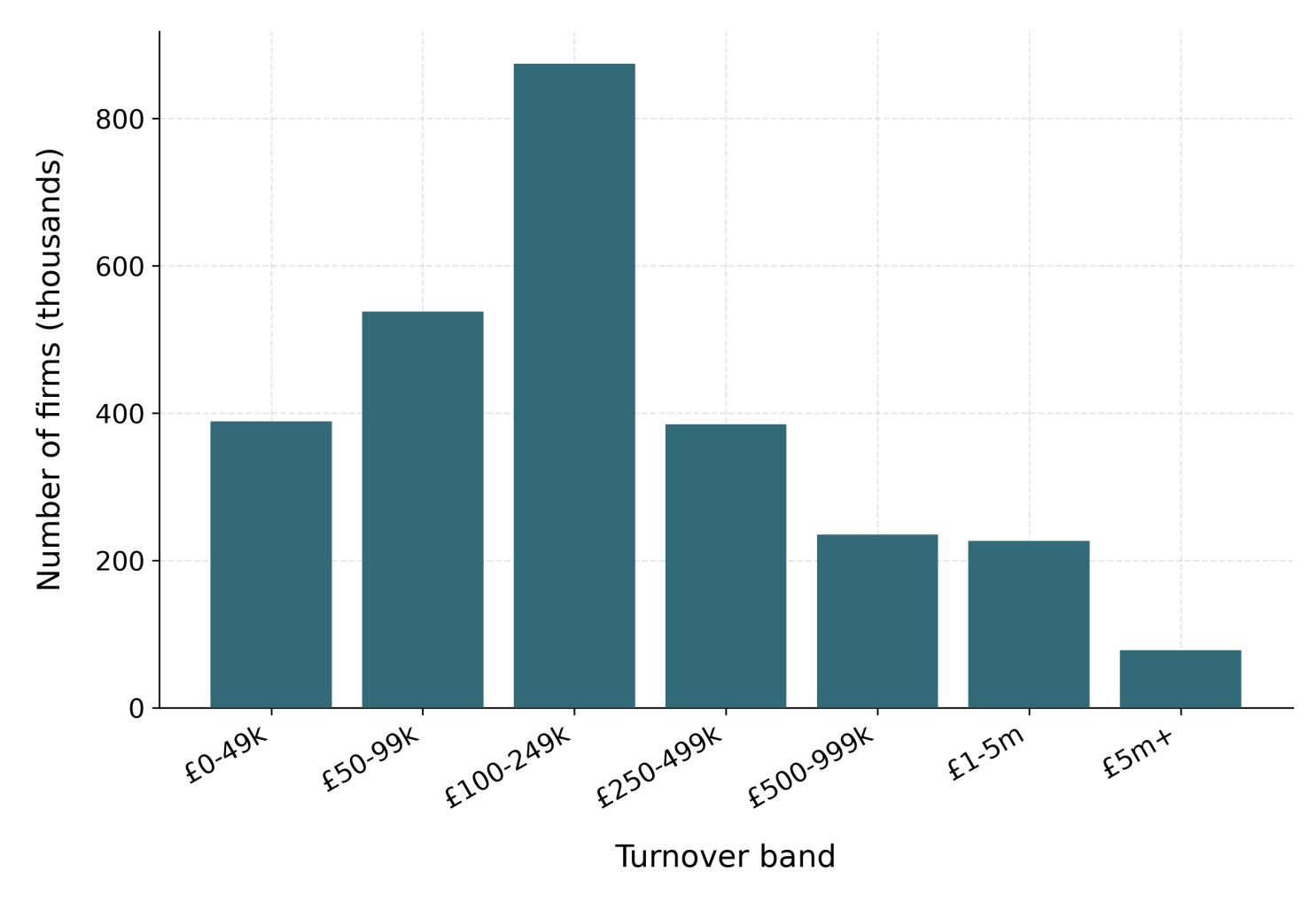




The problem: VAT as a tax *notch*

The UK VAT registration threshold is a **notch**, not a kink. Once taxable turnover crosses T^* , **20% VAT falls due on the whole turnover**, not just the increment — a discrete liability jump of $\tau T^* = \text{£17,000}$ at $T^* = \text{£85,000}$. Threshold **£85,000** (2023–24 data year), raised to **£90,000** in April 2024 — among the **highest in the OECD** — and it sits in a *dense* part of the firm-size distribution. Firms respond by **locating just below** it; the cost of crossing is sharpest for *consumer-facing* firms with little reclaimable input VAT, so the notch distorts both the size distribution and firms' growth at the margin.

VAT/PAYE-registered businesses by turnover band — ONS, 2023–24



Three reform families, one basis

What we cost.

Costed on a *common static basis*:

- **Level** — move the threshold up or down.
- **Shape** — a graduated taper that phases VAT in.
- **Rate** — a reduced-rate band above the threshold.

Contribution

Headline: only the graduated taper removes the notch's dominated region (£550m, vs £497m for a 10% band that removes none); a level rise costs most — and has exactly zero intensive-margin behavioural offset.

An open, reproducible firm-level microsimulation that:

1. costs **level + shape + rate** reforms on one static basis;
2. characterises the notch's distortion via its exact, **elasticity-free dominated region**;
3. proves the near-threshold shape is **target-inherited** (OBR-calibrated; placebo-verified) — in both directions;
4. adds a conditional **behavioural layer** with an exact static nesting.

Data methodology

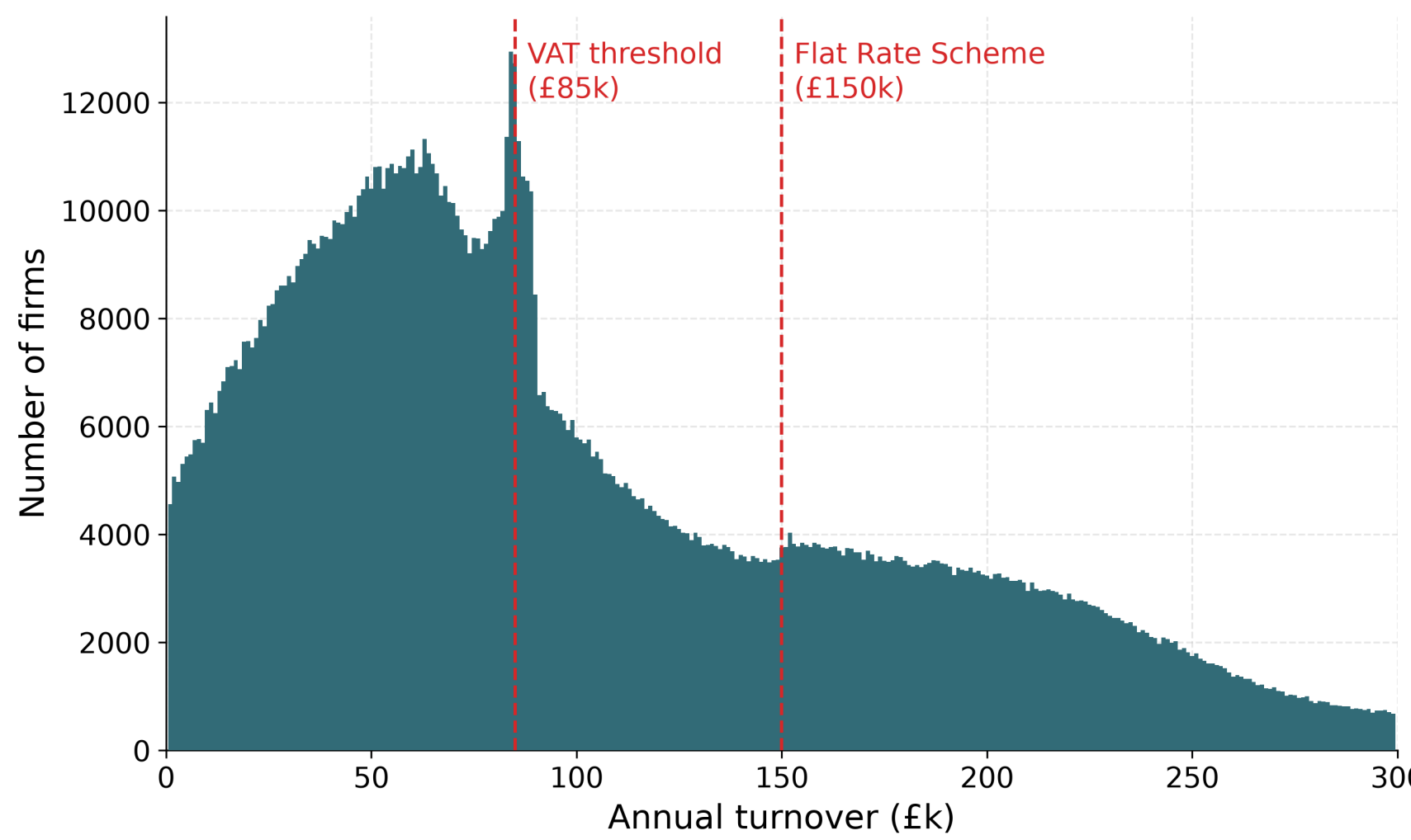
How the synthetic data are built.

Draw synthetic firms by sector $\times$ turnover band; give each an input share $x_i = \rho_i y_i$ and net VAT liability $v_i = 0.20 (y_i - x_i)$ — the standard rate on value added. Firm weights $w_i = e^{\theta_i}$ are fit by gradient descent to HMRC + ONS aggregates, with the near-threshold £1k-band profile pinned to the **OBR's published Chart C data**. The threshold is

Synthetic firm data

→ The engine behind every costing.

$\approx 2.94\text{m}$ firm rows weighted to $\approx 2.5\text{m}$ UK firms, calibrated to HMRC + ONS aggregates (turnover-band counts, sector, employment, net VAT liability by band) at **89.2% / 92.8%** accuracy for the 2023–24 / 2024–25 vintages. Calibrated to the aggregates, so agreement is an **internal-consistency check**, not external validation. Levels are the **VAT/PAYE-registered frame**: unregistered sole traders (most of the all-business cliff at £85k) are outside it.

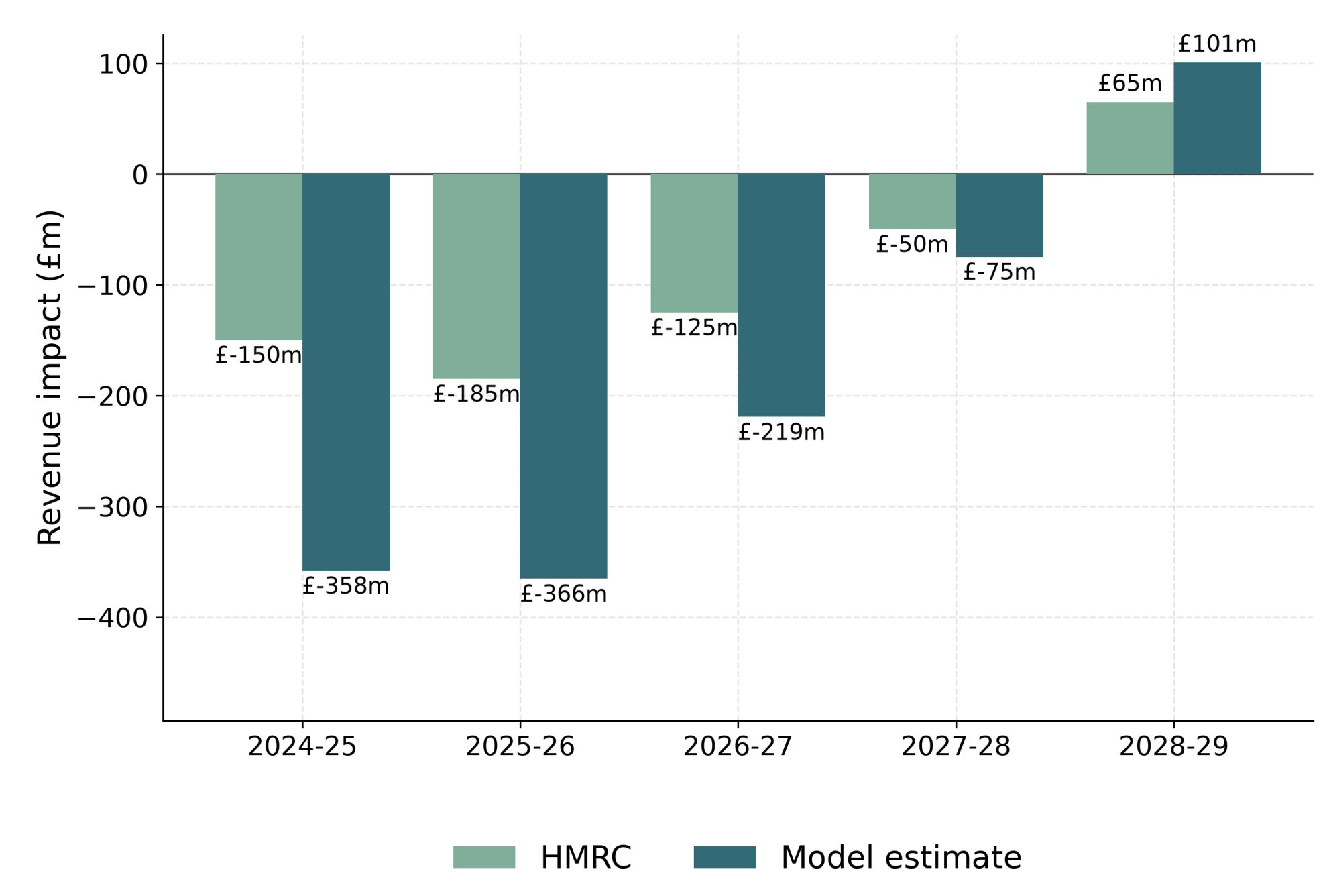


Static costing

Pricing a reform on this data.

$$R(T) = \sum_i w_i v_i \mathbb{1}\{y_i \geq T\}$$

Revenue moves only through firms whose registration status flips. **Anchor vs HMRC:** the £85k→£90k April-2024 rise, under two released-firm conventions, **overshoots early** (–£366m / –£209m retention vs published –£185m in 2025–26 — consistent with HMRC's phased registration response) and **matches by 2026–27** (–£125.1m vs –£125m), with the 2028–29 sign flip reproduced.



Threshold sweep (2025–26)

Scaling the level reform across thresholds.

From a £90k base (**£200.9bn**): roughly **linear**, $\approx \text{£230–265m}$ per

Reform menu

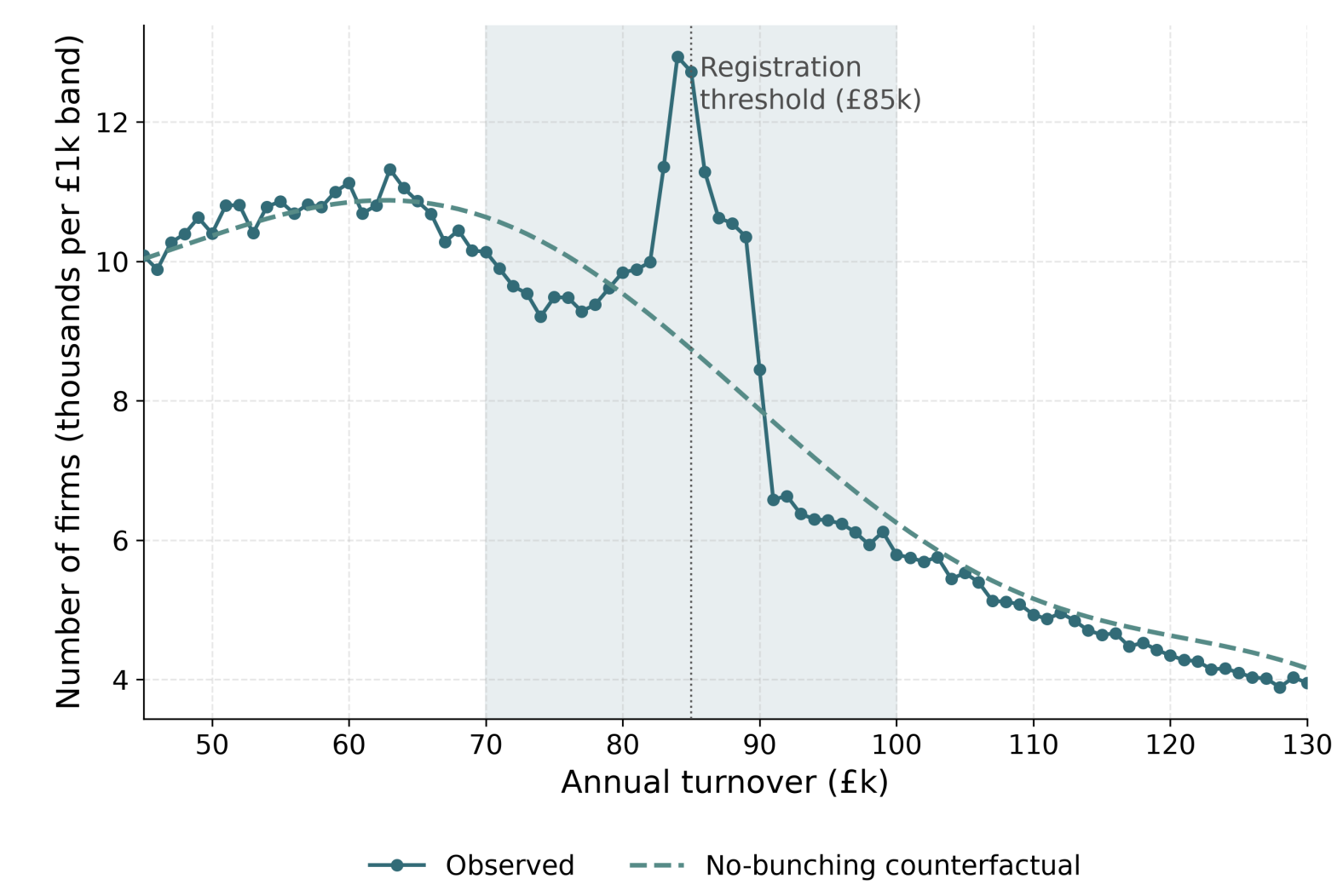
→ Cost and distortion on a common £85k / £184.8bn base.

Reform	Type	Static	Dominated region
Raise to £100k	Level	–£784m	relocated
Taper £85–105k	Shape	–£550m	removed
Reduced 10%	Rate	–£497m	split in two
Reduced 15%	Rate	–£249m	split in two

The near-threshold shape is target-inherited

A transparency result, proved in both directions.

The 2023–24 file reproduces the administrative bunching profile **because it is calibrated to it** (OBR £1k-band targets); the estimator reads back $E = 7,873$. Regenerate *without* those targets $\Rightarrow E = 0$. And with *no* fine targets, the 2024–25 vintage's coarse £90k band edge alone yields $b_{\text{LLAT}} = 1.23$ — within 10% of Liu et al.'s (2021) administrative **1.361**, coincidentally and unstably. Synthetic bunching is **never behavioural evidence**; the administrative estimate remains the behavioural fact.



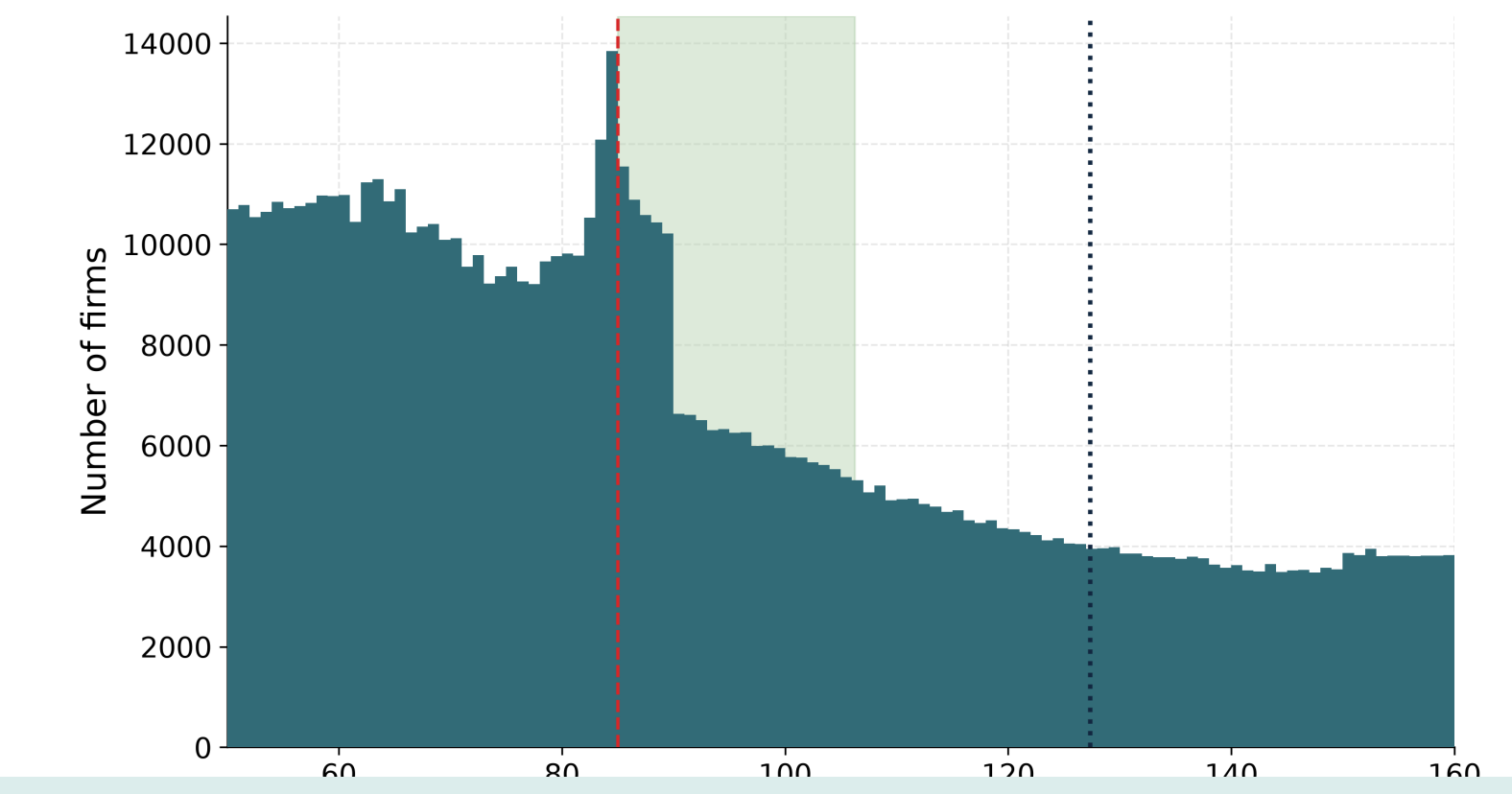
The dominated region

Beyond revenue: measuring the notch's distortion.

Profit jumps *down* by $\tau T^* = \text{£17,000}$ at the threshold. The **dominated region** — turnover where no firm locates because it earns less than at T^* — has exact width

$$a = T^* \frac{\tau}{1-\tau} = \text{£21,250}$$

(£85,000–£106,250), spanning $\approx 160,000$ weighted firms. **Exact, elasticity-free, and input-share invariant** (the deductible share cancels in the value-added formulation).



How reforms change it

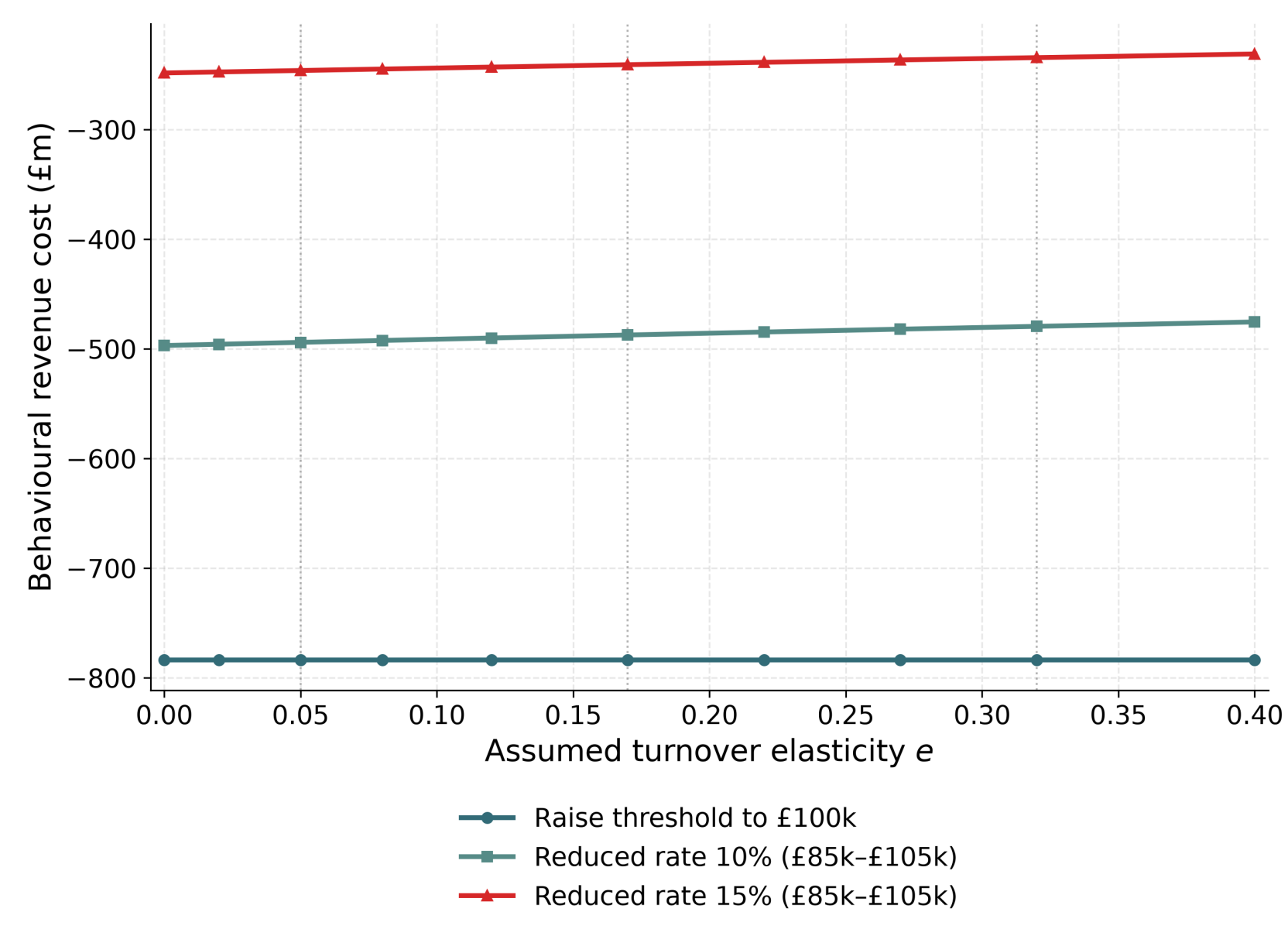
→ Which reforms actually shrink this band?

- Raising the threshold **relocates** the £21,250 band unchanged.
- A banded reduced rate does **not shrink** it. It lowers the *largest single* notch (£17,000→**£12,750** at 15%), but reverting to 20% at the band top adds a *secondary* notch $a' = T_1 \frac{\tau}{1-\tau}$ of £5,250, so total width is **£21,562 (15%) / £22,569 (10%)** — it **splits** the band, not shrinks it.
- **Only the graduated taper removes the distortion** entirely.

Behavioural layer

Layering an assumed response on the static costs.

Conditional on an **assumed** turnover elasticity e (*not* identified on synthetic data); sweep $e \in \{0.05, 0.17, 0.32\}$. The intensive margin **barely moves the costings**: a level rise has **exactly zero** intensive-margin offset at every e (released firms leave the VAT base, so their expansion is untaxed), and the reduced-rate offsets stay **under 4%** (10% band: £497m to **£488m** at $e = 0.17$). The taper is *excluded* — its rate varies continuously, outside the flat-rate machinery. Only $e = 0.05$ has an external anchor, so this is a **sensitivity range**, not a forecast — and the $e \rightarrow 0$ limit reproduces the static costs **exactly**.



Conclusions

What this paper is: an open, reproducible microsimulation that models the UK VAT threshold as a *turnover-tax notch* on synthetic firm data, and prices level, shape and rate reforms three ways — statically, by the exact distortion each removes, and behaviourally.

1. Prices level + shape + rate on one basis; tracks HMRC's anchor costing once the deregistration convention is stated.
2. On the dominated-region metric, **only the taper removes the distortion**; the level move costs most and removes none of it.
3. Sub-band structure in calibrated data is **target-inherited** — never behavioural evidence, in either direction.
4. The intensive margin barely moves the static costings; the static cost and dominated region are the **elasticity-free anchors**.